Your Results, From Zero

For Student 2 · everything explained step by step, with our own numbers

You do not need to be good at statistics to present the Results. You need to understand **one idea per table or figure** and be able to say it simply. This guide teaches you exactly that.

How to use this guide (3 sittings)

1. **Sitting 1 (1.5 h):** Part A (the story of the results) and Part B (statistics lessons). Read slowly. Say every "Say" box out loud.
2. **Sitting 2 (2 h):** Part C (why each table and figure exists, how to read it). Have the thesis open next to you.
3. **Sitting 3 (1.5 h):** Part D (your slides: what to point at), Part E (drills). Then do the drills again the next day without looking.

Keep the one-page cheat sheet (last page) in your pocket for the week before the defence.

Part A · The Results tell one story in nine steps

Every table and figure in Chapter 3 answers one question. They come in a logical order: first **who** answered, then **what they do**, then **how anxious** they are, then the **main link**, then **what else** could explain it, then **who is at risk**, then **behaviour**, then **pathways**, and finally **is it robust?**

Step	Question	Short answer	Where	Slide
1	Who took part?	300 young, educated adults; 44.0% medical students	Table 1	14
2	How do they search?	AI chatbots first (78.0%), search engines second (56.3%)	Table 2, Figure 3	15
3	What do they do after searching?	34.7% delay care when reassured; 31.7% never tell the doctor	Figure 4	22
4	How anxious are they? Do the scales work?	19% elevated; both scales reliable; CSS-12 same in Arabic and English	Tables 3, 4, 5	16
5	Are health anxiety and cyberchondria linked? (H1)	Yes, r = 0.47	Table 6, Figures 5, 6, 11	17, 19
6	Which groups score higher?	Chronic illness, reassurance seekers; AI users (but see step 7)	Tables 7, 8, Figure 7	–
7	Does the link survive other factors? (H2) Who is at risk?	Yes; health anxiety adds 18.4%; AI effect not robust	Tables 9, 10, Figure 8	18, 20, 23
8	Does it affect behaviour? (H4)	Cyberchondria → more disclosure (robust); others exploratory	Table 11	22
9	Is there a cycle? (H3) Is everything robust?	Compatible with a cycle, not proven; all checks unchanged	Table 12, Figures 9, 10, Table 13	21

> **Say:** *Our results answer nine questions in order. In short: AI chatbots are the main source of health information, one in five has elevated health anxiety, and health anxiety and cyberchondria are clearly linked, r = 0.47, even after adjusting for everything else.*

Part B · Statistics from zero (with our numbers)

Each lesson: the idea in plain words, our example, and what to say. Read them in order; each one builds on the one before.

1. Variables: two kinds

- **Categorical** = groups (gender, uses AI yes/no, tool profile). We describe them with **n and %**.
- **Continuous** = scores on a scale (SHAI 0–42, CSS 12–60). We describe them with **mean, SD, median**.

This one fact decides almost every choice of test and chart in the thesis.

2. Mean and SD

Mean = the average. **SD (standard deviation)** = how spread out people are around the average. About two-thirds of people lie within 1 SD of the mean.

```
SHAI mean 12.3 ± SD 5.8 → most people scored between about 6.5 and 18
```

> **Say:** *The average health-anxiety score was 12.3 out of 42.*

3. Median and IQR

The **median** is the middle person. The **IQR** is the range of the middle half. We use them when scores are skewed (not a symmetrical bell shape).

```
SHAI median 12 (IQR 8–16): half of people scored 8 to 16
```

4. Normal distribution and why it matters

Many tests assume scores form a bell curve. The **Shapiro–Wilk test** checks this. Our scores were slightly skewed ($p < 0.01$), so we used **non-parametric** tests (they do not need the bell shape) and **robust** regression.

> **Say:** *Our scores were not perfectly normal, so we used tests that do not assume normality.*

5. p-value

Imagine there is truly **no** link. The p-value is the chance of seeing a result as strong as ours just by luck. Small p (below 0.05) = "unlikely to be luck" = **significant**. It does **not** say how big or important the effect is.

```
r = 0.47, p < 0.001 → almost impossible to get this by chance if there were no
link
```

> **Trap:** p tells you if an effect exists, not how big it is. For size, look at r, β, OR.

6. 95% confidence interval (CI)

Our number is an estimate from 300 people. The CI is the range where the true value probably lies. Narrow CI = precise. If a CI for an odds ratio includes 1, the result is not significant.

```
r = 0.47, 95% CI 0.38–0.55 → the true correlation is probably between 0.38 and 0.55: always clearly above zero
```

7. Correlation r

How closely two scores move together: 0 = no link, 0.1 small, 0.3 moderate, 0.5 large, 1 = perfect. Positive = both go up together.

```
r = 0.47: people with more health anxiety tend to have more cyberchondria. r² = 22%: health anxiety explains about a fifth of the differences in cyberchondria.
```

> **Say:** *Health anxiety and cyberchondria rise together: r = 0.47, a moderate link.*

8. Pearson vs Spearman

Pearson r assumes a straight-line link; **Spearman ρ** uses ranks and does not need normal data. We report both; both were 0.47, so the result does not depend on the method.

9. Partial correlation

The same correlation after "removing" other variables (age, gender, background, chronic disease, trust, AI use). If it stays the same, those variables are not the explanation.

```
r = 0.47 → partial r = 0.48: unchanged
```

> **Say:** *The link stayed the same after adjusting for everything else, so it is not explained by age, gender or background.*

10. Comparing groups: Mann–Whitney and Kruskal–Wallis

To compare a **score** between groups without assuming normality: **Mann–Whitney U** for 2 groups, **Kruskal–Wallis H** for 3 or more.

```
Chronic condition: SHAI 14.2 vs 11.9 → Mann–Whitney p = 0.004
```

11. Chi-square (χ^2)

To test if two **categorical** variables are related, e.g. "elevated health anxiety yes/no" and "uses social media yes/no". It compares the counts we saw with the counts we would expect if there were no link.

12. Many tests → FDR correction (q-value)

If you run 162 tests at p < 0.05, about 8 will be "significant" by pure luck (162 × 0.05 ≈ 8). The **Benjamini–Hochberg FDR** correction adjusts p-values into **q-values**. q < 0.05 = **robust**; p < 0.05 but q ≥ 0.05 = **exploratory** (could be luck).

```
Never disclosing: CSS q = 0.012 (robust) · SHAI q = 0.098 (exploratory)
```

> **Say:** *Because we ran many tests, we corrected for multiple testing; results that did not survive are called exploratory.*

13. Linear regression, B and β

Regression predicts a score (CSS) from several factors at the same time, so each factor is "adjusted" for the others. **B** = how many CSS points change per 1 unit of the factor. **β** = the same on a common scale, so you can compare factors (bigger β = stronger).

```
SHAI: B = 0.68, β = 0.48. Someone with SHAI 20 vs 10 → about 6.8 CSS points
higher.
```

> **Say:** *Each extra point of health anxiety adds about 0.7 points of cyberchondria; it is by far the strongest factor.*

14. R^2, ΔR^2 and hierarchical regression

R^2 = share of differences the model explains (0–1). In **hierarchical** regression we add groups of variables in steps and see how much each step adds (**ΔR^2**).

```
Step 1 demographics R² 0.030 (3%) → step 2 digital use 0.133 (13%) → step 3
health anxiety 0.317 (32%). Health anxiety adds 18.4%.
```

> **Why this was chosen:** Hierarchical order shows that health anxiety matters even after everything else is already in the model.

15. Robust standard errors (HC3)

Regression assumes the "errors" are equally spread. Ours were not (Breusch–Pagan p = 0.004). HC3 robust standard errors fix the p-values and CIs so they are still valid. You only need: "we used robust standard errors because the residuals were uneven".

16. Odds and odds ratios (logistic regression)

Logistic regression predicts a yes/no outcome (elevated health anxiety yes/no). It gives **odds ratios (OR)**: OR 1 = no effect, above 1 = higher odds, below 1 = lower odds. **Adjusted** OR (AOR) = after accounting for the other factors.

```
Worked example from our data:
With elevated health anxiety: 52.6% had high cyberchondria → odds = 52.6/47.4 =
1.11
Without: 23.0% → odds = 23.0/77.0 = 0.30
Odds ratio = 1.11 / 0.30 ≈ 3.7
```

In the model: AOR **3.12** per **1 SD** of cyberchondria (1 SD ≈ 8.4 CSS points) means about three times the odds for every 8 extra points.

> **Trap:** AOR below 1 means protective. Age AOR 0.61: each older age group had lower odds.

17. AUC

How well the logistic model separates people with and without elevated health anxiety. Pick one person with and one without: AUC is the chance the model gives the higher risk to the right one. 0.5 = coin toss, 1 = perfect.

```
AUC 0.83, and 0.79 after correcting for optimism (bootstrap): good
```

> **Why this was chosen:** A model always looks better on the data it was built on, so we corrected the AUC with 500 bootstrap repeats.

18. Bootstrap

A computer trick: draw 5000 new samples of 300 people from our own data (with replacement), redo the analysis each time, and see how much the result varies. That gives a CI without assuming normality. We used it for mediation and to check the regression coefficient.

19. Mediation (indirect effect)

Does part of the link from X to Y go **through** a middle variable M? Path **a** = X → M; path **b** = M → Y (adjusted for X); **indirect** = a × b; **direct** (c′) = what is left.

```
Model A: health anxiety → excessive searching → distress
a = 0.46, b = 0.35 → indirect = 0.46 × 0.35 ≈ 0.16
total = 0.46, direct = 0.30 → about 35% goes through excessive searching
Model C: health anxiety → distress → reassurance seeking: indirect 0.22, direct
0.01 → almost all goes through distress
```

> **Trap:** Mediation in one survey is NOT proof of cause. The reverse order was also significant (0.15). Always say: compatible with a cycle, not proof.

20. Moderation (interaction)

Does a third variable change the **strength** of the link? E.g. is the SHAI–CSS link stronger in AI users? We tested 7 moderators; none were significant after FDR. So the link is similar in everyone.

21. Reliability: Cronbach α and McDonald ω

Do the items of a scale measure the same thing? 0.7 acceptable, 0.8 good, 0.9 excellent.

```
SHAI α 0.81, CSS α 0.86: good
```

22. CFA and fit indices

Confirmatory factor analysis checks whether the items group into the factors the scale claims (CSS-12 = four facets). Fit indices: **CFI** near 0.95 or higher = good; **RMSEA** and **SRMR** 0.08 or lower = acceptable.

```
CSS-12 four factors: CFI 0.95, RMSEA 0.070, SRMR 0.053 → acceptable · SHAI one
factor: CFI 0.83 → poor (it has two factors)
```

23. Measurement invariance

Before comparing Arabic and English answers, we must check the questionnaire works the same in both, like checking two rulers measure the same centimetre. Three steps: **configural** (same shape), **metric** (items equally strong), **scalar** (same starting points). If fit does not get worse at a step (CFI drop ≤ 0.01), that step passes.

CSS-12 passed all three. SHAI-14 passed the first two.

24. DIF

Differential item functioning: does a **single question** behave differently in Arabic vs English for people with the same total score? None did after FDR correction.

25. Sensitivity analysis

Re-doing the main analysis in another way (excluding doubtful responses, removing one item, another method) to see if the result changes. Ours did not change: r stayed 0.46–0.48.

Part C · Every table and figure: why it is there and how to read it

The rules that decided every choice

If the data are...	...we used	Example
Exact numbers that people need to look up	a **table**	Table 1 participants, Table 9 regression
A pattern people should see at a glance	a **figure**	Figure 5 scatter plot
Percentages of groups (categorical)	**bar chart**	Figures 3, 4
Two continuous scores together	**scatter plot**	Figure 5
Many correlations at once	**heatmap** (colour = strength)	Figures 6, 11
A score compared across groups, skewed data	**box plot** (median, spread, outliers)	Figure 7
Means of several facets in two groups	**bar chart with 95% CI**	Figure 10
Many odds ratios with CIs	**forest plot**	Figure 8
Paths between variables	**path diagram**	Figure 9

Also: the most important results get **both** a table (exact numbers) and a figure (the picture), e.g. the main correlation (Table 6 + Figure 5) and the logistic model (Table 10 + Figure 8). Less central results have only a table.

Table 1 · Who took part

> **Why this was chosen:** Every study must describe its sample first, so the reader knows who the results apply to. A table, because the reader needs exact n and %.

How to read it:

1. Rows = characteristics (age, gender, education, study/work, background, chronic disease, language).
2. Columns = n (number of people) and %.

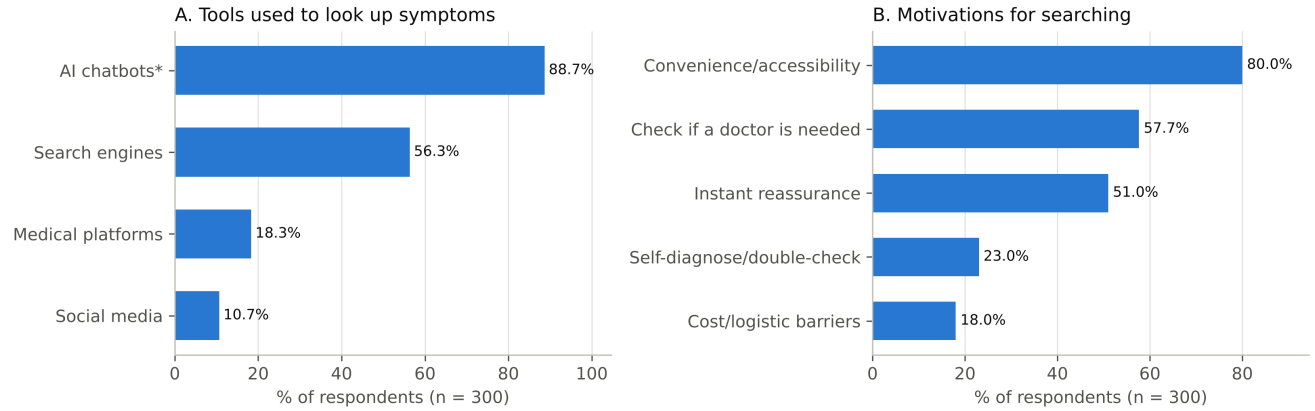
What it shows: A young sample (62.3% aged 20–29), 44.0% medical students, 63.3% with a health background.

> **Say:** *Our 300 participants were mostly young and educated; almost half were medical students. This matters when we interpret percentages.*

> **Trap:** Why so many medical students? Answer: the link spread through our own networks first; we say percentages cannot be generalised.

Table 2 + Figure 3 · How people search and why

> **Why this was chosen:** Tools and motives are categories with multiple answers, so percentages are best shown as bars (Figure 3); Table 2 keeps every exact count.



How to read it:

1. Figure 3, panel A = tools; panel B = motives.
2. Each bar = % of all 300 who ticked that option.
3. Bars add up to more than 100% because people could tick several.

What it shows: AI chatbots 78.0% ticked (88.7% including those who named one) vs search engines 56.3%; convenience is the top motive.

> **Say:** *AI chatbots are now the main gateway: 78.0% ticked them, compared with 56.3% for search engines.*

> **Trap:** Why two AI numbers? 32 people named a chatbot but did not tick "AI"; we report both and test both.

Figure 4 · What people do after searching

> **Why this was chosen:** Three related sets of categories (behaviour, disclosure, relationship) shown side by side so the committee sees the whole consultation story in one picture.



How to read it:

1. Panel A: behaviours after searching (multiple answers).
2. Panel B: do they tell the doctor? (always / sometimes / never).
3. Panel C: effect on the relationship with doctors.

What it shows: 34.7% delay care when reassured; 31.7% never tell the doctor; 34.7% say trust was strained.

> **Say:** *Searching changes the consultation: one in three delays care because they feel reassured, and almost one in three never tells the doctor.*

Table 3 · Scores and reliability

> **Why this was chosen:** Before using any score, we must show the scales work (reliability) and give their levels. A table because these are exact values.

How to read it:

1. Rows: SHAI, CSS and the four CSS facets.
2. Columns: mean ± SD, median (IQR), range, α (95% CI), ω.
3. The note under the table has the CFA fit.

What it shows: Both scales reliable (α 0.81 and 0.86); excessiveness is the highest facet, compulsion the lowest.

> **Say:** *Both questionnaires were reliable. People search a lot, but it rarely disrupts their daily life.*

Table 4 · Prevalence at different cut-offs

> **Why this was chosen:** The % with "elevated" health anxiety depends on the cut-off you choose. Showing four cut-offs is honest and stops the committee from accusing us of choosing the one that looks best.

How to read it:

1. Each row = a cut-off (15, 18, 22, 29).
2. Columns: n, %, 95% CI.
3. Bold row = 18, the one we use.

What it shows: 32.7% at 15, 19.0% at 18, 7.3% at 22, 0.7% at 29.

> **Say:** *About one in five scored 18 or more; this is a screening threshold, not a diagnosis, and the table shows how the estimate changes with the cut-off.*

> **Trap:** Why 18? It best separated people with health anxiety from healthy controls in the Swedish study.

Table 5 · Arabic vs English (measurement invariance)

> **Why this was chosen:** We combined Arabic and English answers. To do that fairly, we had to prove the questionnaire works the same in both languages.

How to read it:

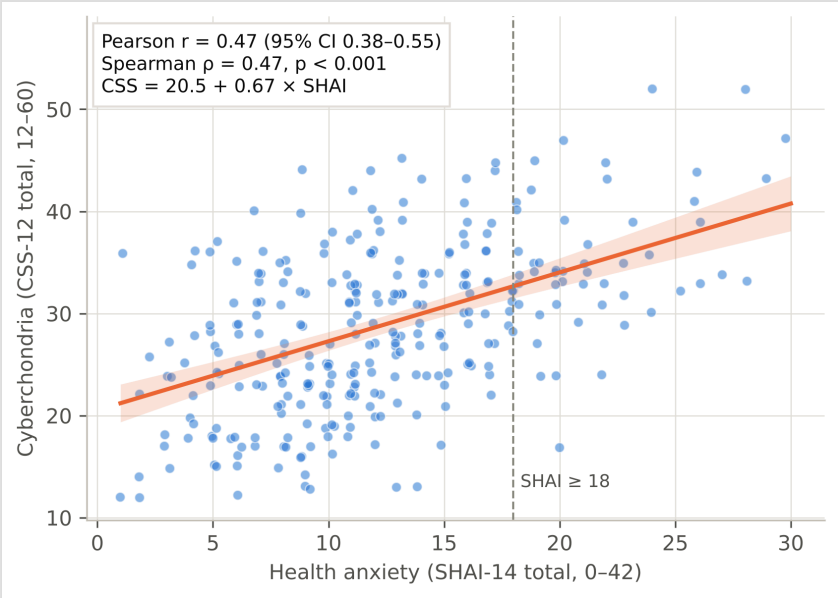
1. Two blocks: CSS-12 and SHAI-14.
2. Three rows each: configural, metric, scalar.
3. Look at ΔCFI and the Decision column: "Supported" = that step passed.

What it shows: CSS-12 passed all steps; SHAI-14 passed metric, not scalar; no item behaved differently.

> **Say:** *The cyberchondria questionnaire works identically in Arabic and English, so combining the two languages was justified.*

Table 6 + Figure 5 · The main result

> **Why this was chosen:** This answers our main question (H1), so it gets both: a figure to see the pattern, a table for exact numbers. A scatter plot is the standard picture for two continuous scores.



How to read it:

1. Each dot = one participant (x = SHAI, y = CSS).
2. The red line = best-fit line; the shaded band = its 95% CI.
3. The dotted vertical line = SHAI 18 cut-off.
4. Table 6: r for the total and for each facet, plus adjusted β.

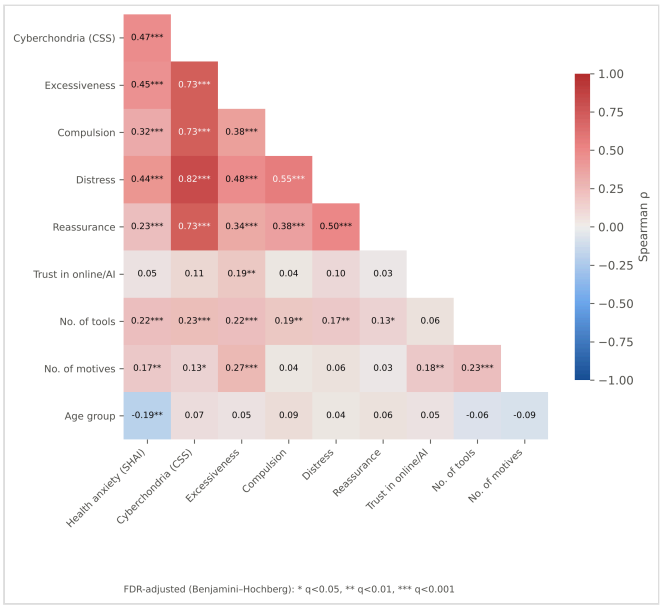
What it shows: Clear upward trend: r = 0.47; strongest for excessiveness (0.46) and distress (0.45).

> **Say:** *This is our main result: the more health anxiety, the more cyberchondria, r = 0.47, and it holds after adjustment.*

> **Trap:** Point at the line going up, then at the r value in the box. Do not say "causes".

Figure 6 · Correlation heatmap

> **Why this was chosen:** We had 10 variables. Writing 45 correlations in text is unreadable; a heatmap shows them all at once, with colour for strength.



How to read it:

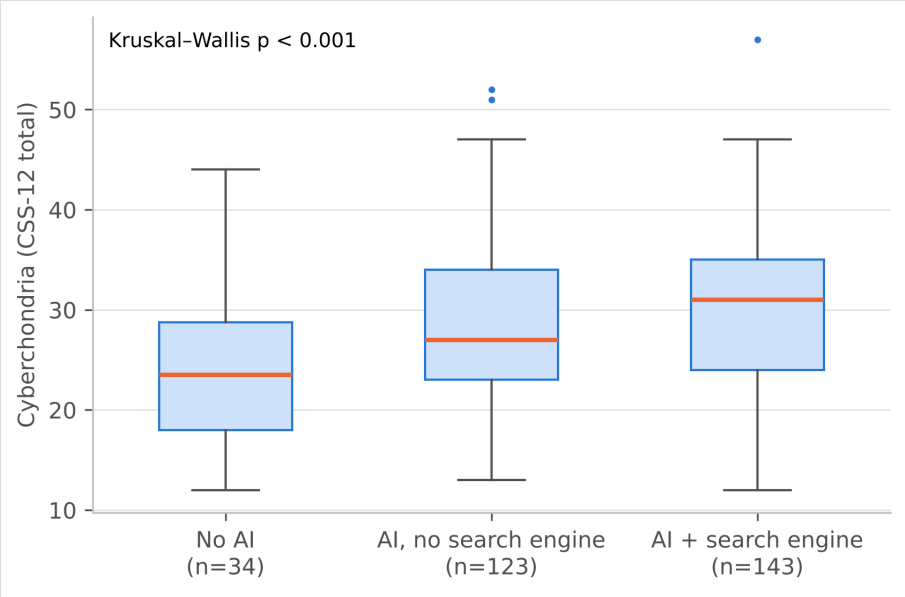
1. Each square = correlation between the row and column variable.
2. Darker red = stronger positive; blue = negative.
3. Stars = significant after FDR (* q<0.05, ** q<0.01, *** q<0.001).

What it shows: Health anxiety correlates with every cyberchondria facet; trust relates only to excessiveness; age is slightly negative.

> **Say:** *The heatmap shows all relationships at once: health anxiety is linked with every part of cyberchondria.*

Table 7 + Figure 7 · Which groups score higher

> **Why this was chosen:** Table 7 compares the scores across every participant group (exact means and p/q). Figure 7 is a box plot because CSS is skewed: it shows the median and spread, not just the mean.



How to read it:

1. Box plot: middle line = median; box = middle half of people; whiskers = range; dots = outliers.
2. Three groups: no AI, AI only, AI + search engine.

What it shows: Health anxiety higher with chronic illness (14.2 vs 11.9) and in reassurance seekers (13.8 vs 10.9); CSS higher in AI users (29.5 vs 24.0), highest in AI + search engine users.

> **Say:** *People who combine AI with search engines had the highest cyberchondria, but we will see this is not robust once we adjust.*

> **Trap:** Do not over-sell the AI difference; it disappears in the strict analysis.

Table 8 · Yes/no outcomes (χ^2)

> **Why this was chosen:** Some outcomes are yes/no (elevated health anxiety, high cyberchondria). χ^2 tests show which categories go with them. Only significant ones are listed to keep it short.

How to read it:

1. Each row: variable, % with the outcome in each group, Cramér's V (strength), p, q.

What it shows: With elevated health anxiety, 52.6% had high cyberchondria vs 23.0% without (OR ≈ 3.7).

> **Say:** *People with elevated health anxiety were much more likely to have high cyberchondria.*

Table 9 · What predicts cyberchondria (regression)

> **Why this was chosen:** Group comparisons look at one factor at a time; regression tests all factors together, so we know which matter independently (H2).

How to read it:

1. Rows = predictors; B = CSS points per unit; β = comparable strength; p.
2. Note under the table = R^2 for each step.

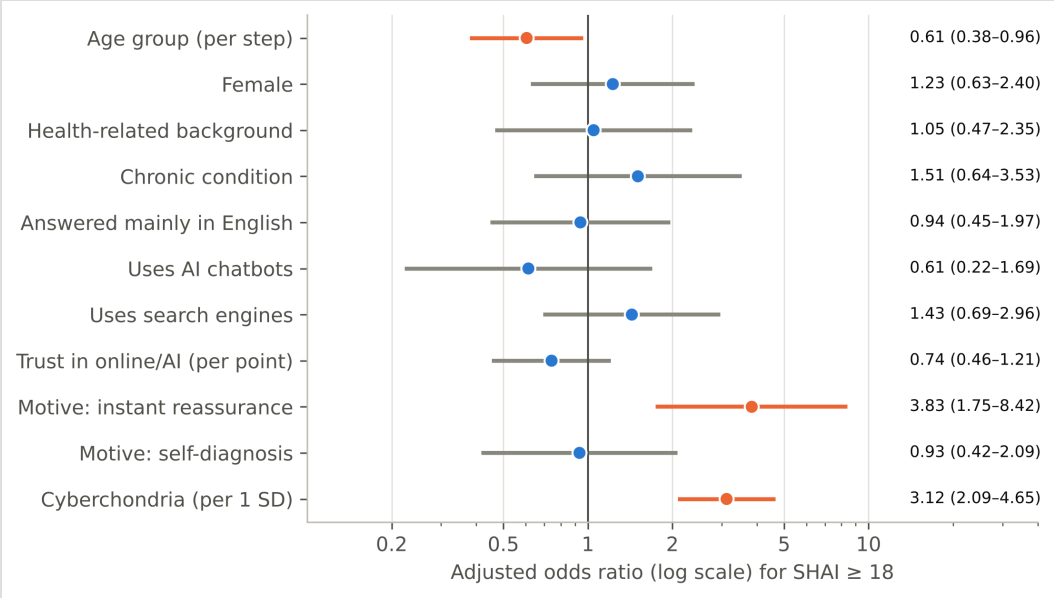
What it shows: Health anxiety β 0.48 (largest by far); R^2 rises from 0.030 to 0.317.

> **Say:** *Demographics explain only 3%; adding health anxiety explains a further 18.4%. Health anxiety is by far the strongest predictor.*

> **Trap:** AI row: significant here, but not with the strict definition (slide 23).

Table 10 + Figure 8 · Who has elevated health anxiety

> **Why this was chosen:** For a yes/no outcome we use logistic regression; a forest plot is the standard way to show many odds ratios with CIs at once.



How to read it:

1. Each dot = AOR; the line = its 95% CI.
2. Vertical line at 1 = no effect. If the line crosses 1 → not significant.
3. Log scale: 0.5 and 2 are the same distance from 1.
4. Orange = significant.

What it shows: Cyberchondria AOR 3.12, reassurance motive 3.83, age 0.61 (protective); AUC 0.83 → 0.79.

> **Say:** *Three factors stand out: higher cyberchondria, searching for reassurance, and younger age.*

Trap: Only 57 events, so the reassurance odds ratio has a wide CI; say “exploratory”.

Table 11 · Behaviour models

Why this was chosen: To test H4 we need adjusted models for each behaviour. A table because there are 8 outcomes × 2 scores: too many for a chart.

How to read it:

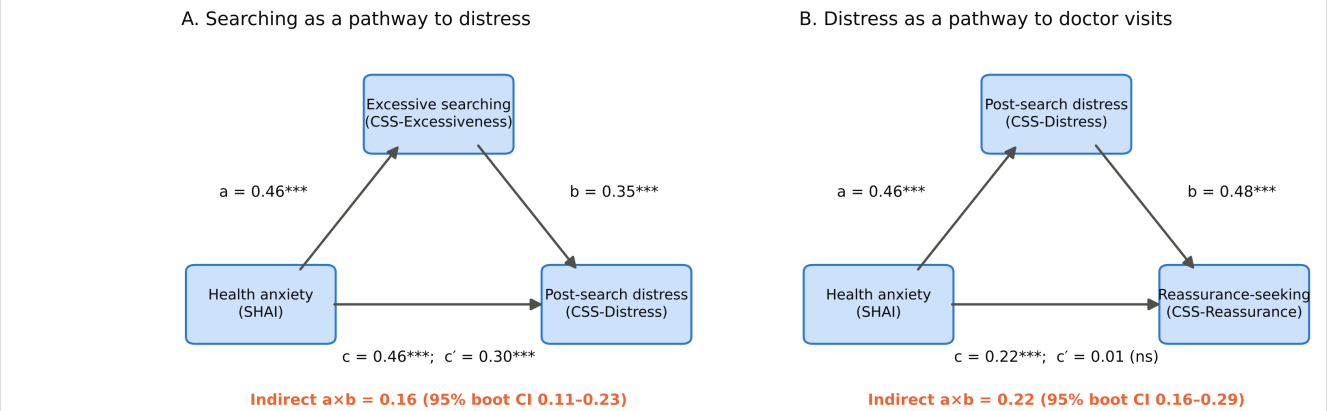
1. Each row = a behaviour; columns = AOR (95% CI) and p for SHAI and for CSS.

What it shows: Higher CSS → urgent visits (AOR 1.51), less ignoring (0.65), less never-disclosing (0.57, the only robust one).

Say: *Cyberchondria, not health anxiety, is what goes with behaviour; only the link with disclosure survived correction.*

Table 12 + Figure 9 · Pathways (mediation)

Why this was chosen: To test H3 (the cycle). A path diagram is the clearest way to show a, b and direct paths; the table adds the CIs and the reverse model.



Standardised coefficients adjusted for age, gender, health background, chronic illness, trust and AI use; 5,000 bootstrap resamples. * p<0.05, ** p<0.01, *** p<0.001.

How to read it:

1. Boxes = variables; arrows = paths with coefficients.
2. a = X → M, b = M → Y, c′ = direct.
3. Below each diagram: the indirect effect with its CI.

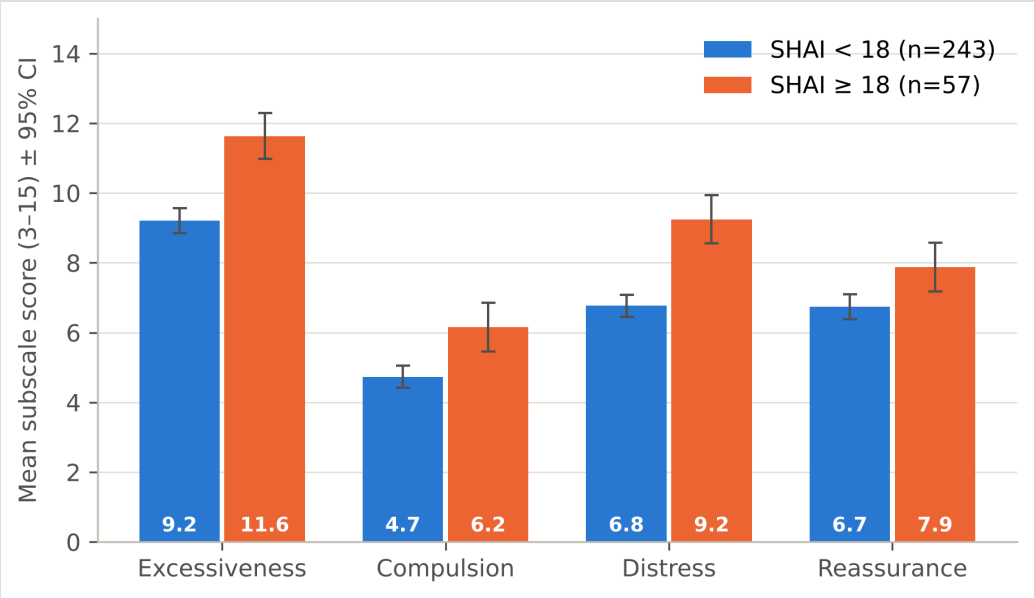
What it shows: Via excessive searching 0.16; via distress 0.22 (direct falls to 0.01); reverse also 0.15.

Say: *The pattern fits a cycle, but because the reverse direction also fits, we cannot say which comes first.*

Trap: Never say “proves”.

Figure 10 · Facets by health-anxiety group

Why this was chosen: Shows simply that people with elevated health anxiety score higher on every facet, and which facets differ most. Bars with 95% CI are best for comparing means of two groups.



How to read it:

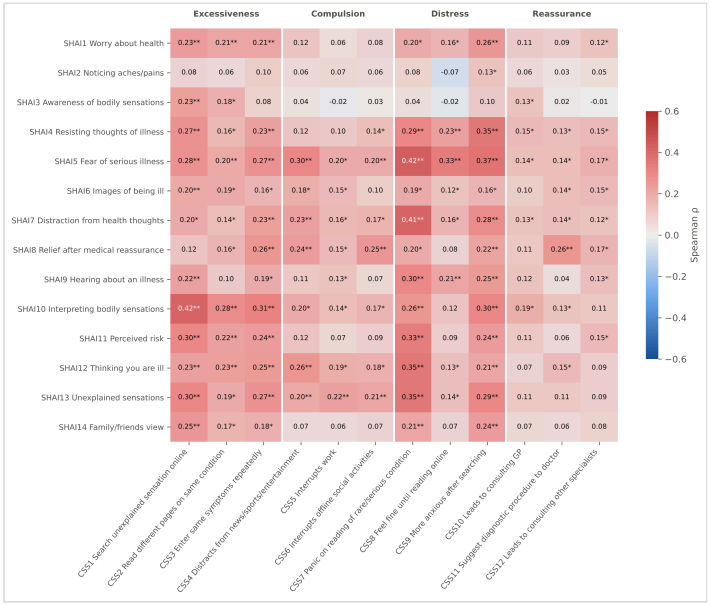
1. Blue = SHAI < 18 (243 people); orange = SHAI ≥ 18 (57 people).
2. Bar height = mean; small lines = 95% CI.

What it shows: Biggest gaps: excessiveness (9.2 vs 11.6) and distress (6.8 vs 9.2).

> **Say:** *People with elevated health anxiety score higher on every part of cyberchondria, most of all on excessive searching and distress.*

Figure 11 (Appendix) · Item by item

> **Why this was chosen:** To see which exact questions drive the link (168 pairs). Only a heatmap can show 168 numbers readably, so it goes in the appendix.



How to read it:

1. Rows = 14 SHAI items; columns = 12 CSS items.
2. Colour = strength; stars = significant after FDR (**: also after the stricter Holm correction).

What it shows: 111/168 pairs significant; strongest: fear of serious illness ↔ panic on reading of a rare condition (0.43). Noticing aches is almost unrelated.

> **Say:** *What drives searching is fear and interpretation, not simply noticing symptoms.*

Table 13 · Sensitivity analyses

> **Why this was chosen:** Committees always ask “is it robust?”. One table answers all such questions at once.

How to read it:

1. Each row = a check; Result; Conclusion column.

What it shows: r stays 0.46–0.48 whatever we exclude; only the AI effect depended on a decision.

> **Say:** *Every check left the main result unchanged.*

Part D · Your 11 slides: what to point at and what to say

Short versions (three sentences each). The full scripts are in the speaker notes and in your Student 2 guide. Aim for 30–35 seconds per slide.

Slide	Point at	Say (short version)
13 · Results divider	Nothing to point at.	Thank you. I will present our results in six steps: who took part, how they search, the levels of health anxiety and cyberchondria, the main association, the predictors and pathways, and behaviour and AI.
14 · Participants (Table 1)	The highlighted rows: gender and medical students.	Table 1 shows our 300 participants. Most were young, 62.3% aged 20 to 29, and 44.0% were medical students. In summary, a young and educated sample, which we keep in mind when reading percentages.
15 · How people search (Figure 3)	The top bar of panel A (AI chatbots), then the four number boxes.	Figure 3 shows the tools people use. AI chatbots come first, 78.0%, ahead of search engines at 56.3%, and ChatGPT alone is used by 59.3%. In summary, for many people the first “doctor” is now an AI chatbot.
16 · Levels & reliability (Table 3)	The first two rows, then the 19% box.	Table 3 shows the scores. 19% scored in the elevated range of health anxiety, and both questionnaires were reliable, with alpha 0.81 and 0.86. In summary, our tools measured well, in both Arabic and English.
17 · Main result (Figure 5)	The rising line, then the r box.	This is our main result. Health anxiety and cyberchondria rise together, r = 0.47, and the link stays 0.48 after adjusting for everything else. In summary, hypotheses 1 and 2 are supported.
18 · What predicts cyberchondria (Table 9)	The R^2 rows, then the 18.4% box.	Table 9 is our regression. Demographics explain only 3% of cyberchondria, but health anxiety adds another 18.4%. In summary, how anxious you are matters far more than who you are.
19 · Item by item (Figure 11)	The 0.43 and 0.42 boxes, then the “≈ 0” box.	Figure 11 looks at every pair of questions. The strongest links are fear of serious illness and interpreting body sensations; simply noticing aches is almost unrelated. In summary, fear drives searching, not noticing.
20 · Who has elevated health anxiety (Figure 8)	The orange dots, then the four boxes.	Figure 8 shows the odds ratios. Higher cyberchondria triples the odds, searching for reassurance nearly quadruples them, and older age is protective; the model's AUC is 0.79 after correction. In summary, the person most at risk is young and searches for reassurance.
21 · Pathways (Figure 9)	Panel A indirect effect, then panel B direct effect, then the orange box.	Figure 9 shows the pathways. Part of the link runs through excessive searching (0.16), and almost all of the link to reassurance seeking runs through distress. In summary, this is compatible with a cycle, but one survey cannot prove the direction.
22 · Behaviour (Figure 4)	The first bar of panel A, then the four boxes.	Figure 4 shows what people do after searching: 34.7% delay care because they feel reassured, and 31.7% never tell their doctor. Higher cyberchondria goes with more disclosure, and that result survived correction. In summary, searching changes the consultation before it begins.
23 · AI chatbots	The table: inclusive vs strict rows.	Finally, AI. AI users scored 5.3 points higher, but with the strict definition the difference was only 1.5 and not significant, p = 0.31. In summary, there is no reliable independent effect of AI. I now hand over to my colleague.

Part E · Drills: explain it like you would to a 12-year-old

Cover the right column. Answer out loud in one or two sentences. Then check.

Explain...	Model answer
What is a p-value?	The chance of getting a result this strong by luck if there were really no effect. Below 0.05 we call it significant.
What does r = 0.47 mean?	When health anxiety goes up, cyberchondria tends to go up too; a moderate link, explaining about a fifth of the differences.
What is a confidence interval?	The range where the true value probably lies. Ours for r is 0.38 to 0.55.
Why did you correct for multiple tests?	With 162 tests, about 8 would be significant just by luck; FDR correction separates real results from probable luck.
What is regression for?	To see which factors matter when we look at all of them together.
What is β?	The strength of each factor on a common scale, so we can compare them. Health anxiety had the biggest, 0.48.
What is an odds ratio of 3?	The odds are three times higher. Below 1 means lower odds.
What is AUC?	How well our model tells anxious from non-anxious people apart: 0.79 is good.
What is mediation?	Checking whether part of a link passes through a middle step, here searching or distress.
Does mediation prove a cycle?	No. The reverse also fits, and it is one survey at one time point.
Why did the AI effect disappear?	It depended on 32 people who named a chatbot but did not tick the box; only 34 non-users. With the strict definition it was not significant.
Why non-parametric tests?	Our scores were not perfectly normal; these tests do not need that.
What is Cronbach's alpha?	Whether the questions of a scale measure the same thing; ours were 0.81 and 0.86, good.
What is measurement invariance?	Checking the questionnaire measures the same thing in Arabic and English, like checking two rulers. The CSS-12 passed fully.
What is a sensitivity analysis?	Doing the analysis in other ways to see if the result changes. It did not.
Why both a table and a figure for some results?	The figure shows the pattern; the table gives the exact numbers. We did both for the most important results.
Why a box plot for Figure 7?	Cyberchondria scores are skewed, so the median and spread are more honest than a bar of means.
Why a forest plot for Figure 8?	It shows many odds ratios and their CIs at once; anything crossing 1 is not significant.
Why a heatmap for Figure 6?	There are 45 correlations; colour lets you see them all at once.
Why is Figure 11 in the appendix?	It has 168 numbers: important detail, but too much for the main chapter.

Cheat sheet (one page)

What	Number	Where
Participants	300; 62.3% aged 20–29; 44.0% medical students	Table 1
AI chatbots / search engines	78.0% (88.7% incl. named) / 56.3%; ChatGPT 59.3%	Figure 3
Delay care when reassured / never tell the doctor	34.7% / 31.7%	Figure 4
SHAI mean / CSS mean	12.3 / 28.9	Table 3
Reliability α	0.81 / 0.86	Table 3
Elevated health anxiety	19% (CI 15.0–23.8%); 32.7 / 19.0 / 7.3 / 0.7% at 15/18/22/29	Table 4
Invariance	CSS-12 scalar ✓; SHAI-14 metric ✓, scalar ×; no DIF	Table 5
Main correlation	r = 0.47 (0.38–0.55); partial 0.48; r^2 22%	Table 6, Figure 5
Facets	excess 0.46, distress 0.45, compulsion 0.25, reassurance 0.24	Table 6
Regression R^2	0.030 → 0.133 → 0.317; ΔR^2 18.4%; β 0.48; B 0.68	Table 9
Logistic	AOR 3.12 / 3.83 / 0.61; AUC 0.83 → 0.79	Table 10, Figure 8
Behaviour (robust)	CSS → never disclose AOR 0.57, q 0.012	Table 11
Mediation	via excess 0.16 (0.11–0.23); via distress 0.22, direct 0.01; reverse 0.15	Table 12, Figure 9
AI effect	+5.3 inclusive vs +1.5 strict (p 0.31); 32 named-not-ticked; 34 non-users	Table 9, slide 23
Item level	111/168 FDR; top ρ 0.43	Figure 11
Robustness	r 0.46–0.48 in every check	Table 13

> **Say:** *Health anxiety and cyberchondria travel together, r = 0.47; AI chatbots are the main source but have no reliable independent effect; and searching changes the consultation before it begins.*